

Material Behavior Prediction

Milestone 8

Deployment Ready

Complete Technical Guide to Milestone 8: New Material Prediction Interface & Model Persistence

1. The Goal: From Trained Model to Usable Engineering Tool

The final stage of practical machine learning is packaging the trained algorithm into a clean, reusable function that accepts new material formulations and returns predictions with domain validation and structural grading:

The Engineering Interface Paradigm

```
predict_material(cement, slag, ash, water, superplastic, coarseagg, fineagg, age) →  
Strength (MPa)
```

2. The `predict_material()` Pipeline Architecture

Implemented in `src/predict_new_material.py`, the function performs 5 automated steps:

1 Physical Boundary & Sanity Validation

Rejects negative masses and zero curing durations. Checks total wet density (approx 2,200 to 2,500 kg/m³) and warns if the curing age exceeds the 365-day training boundary.

2 Exact Schema Formatting

Converts input variables into a 1-row pandas DataFrame matching the exact feature names and data types used during model training.

3 Champion Model Inference

Loads `models/final_material_model.joblib` (our 92.0% R^2 Gradient Boosting model) and generates the continuous prediction in Megapascals (MPa).

4 Materials Engineering Indicators & Grade Classification

Computes key chemistry ratios and categorizes the predicted strength into standard structural civil engineering grades:

- **< 20 MPa:** Lean Concrete (Sub-bases, non-structural paving, mass filling).
- **20 – 35 MPa:** Standard Structural Concrete (Residential foundations, slabs, columns).
- **35 – 50 MPa:** Heavy Commercial Structural Concrete (Multi-story buildings, highway bridges).
- **> 50 MPa:** High-Performance Concrete (Pre-stressed beams, skyscrapers, marine structures).

5 **Mandatory Scientific Disclaimer**

Explicitly prints that the output is an algorithmic statistical estimate, not a physical destructive test, preserving engineering safety standards.

3. Verification Across 3 Real-World Engineering Scenarios

Scenario	Formulation Details	Key Ratio	Predicted Strength	Engineering Structural Grade
Case 1: Standard Residential	Cement 300 kg, Water 180 kg, 0 slag, 0 ash, Age 28 days	\$w/b = 0.60\$	31.31 MPa	Standard Structural Concrete (Meets 25–30 MPa building code for slabs and footings).
Case 2: Green High-Performance	Cement 200 kg, Slag 120 kg, Ash 80 kg, Water 150 kg, SP 10 kg, Age 56 days	\$w/b = 0.38\$	57.08 MPa	High-Performance Concrete (Exceeds 50 MPa while cutting clinker carbon emissions by 45%!).
Case 3: Rapid Early-Strength Repair	Cement 450 kg, Slag 50 kg, Water 140 kg, SP 15 kg, Age 3 days	\$w/b = 0.28\$	38.39 MPa	Heavy Commercial Concrete (Attains 38+ MPa in just 72 hours for fast highway opening).

How to Use `predict_material` in Any Python Script

```
from predict_new_material import predict_material

strength = predict_material(
    cement=350.0, water=160.0, slag=100.0, ash=0.0,
    superplastic=8.0, coarseagg=1000.0, fineagg=750.0, age=28
)
print(f"Predicted Strength: {strength:.2f} MPa")
```